NED University of Engineering and Technology
Department of Software Engineering

GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Search, Screening, Quality Assessment and Data Extraction

Operating Systems CEP, Topic 23
Task 2: Working Evidence Report through Guideline Section 14

Eshaal Faisal (SE-24008)
Raazia Imran Reshamwala (SE-24009)
Rameen Siddiqui (SE-24016)

Interim evidence report. Preliminary recommendations are complete for the active record queue. Independent validation, final eligibility and the final included-study count remain pending. Later CEP synthesis is outside this cycle.

16 October 2026 (Pakistan time)

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1 Background

Concurrent inference tenants can compete for compute, memory capacity, memory bandwidth and execution opportunities on one physical GPU. Sharing may increase useful work, but a change in throughput does not establish predictable request latency or resource and fault isolation. The review examines the assigned families: time-slicing, MPS, MIG and API interception/remoting. Mechanism definitions follow the cited documentation; numerical conclusions require experimental primary studies.

Context definitions: [1, 2, 3]. The submitted 27 September protocol is preserved unchanged.

2 Engineering problem and scope

The engineering problem is to choose a sharing mechanism under specified inference-serving requirements, hardware constraints and latency/isolation priorities. The unit of interest is concurrent inference on one physical GPU. Distributed systems are eligible only where an extractable within-device inference-sharing result can be separated. Remote execution alone is insufficient: the remote endpoint must be shared and tenant effects evaluable. Mixed training/ inference studies require separable inference outcomes.

3 Objective and research questions

Collect and appraise evidence that can support a conditional engineering decision. This cycle strengthens the evidence registers; it does not select a universal winner or answer the final CEP synthesis questions.

ID	Preserved protocol question
RQ1	Which evaluated sharing families, granularities, implementation layers and isolation boundaries can be classified?
RQ2	How are utilization, throughput, density, latency and interference measured, and how comparable are methods?
RQ3	Under which GPU, workload, batch, tenant mix and configuration conditions do outcomes change?
RQ4	What resource/fault isolation, costs, trade-offs and limitations are supported?
RQ5	Which conditional engineering choices are supported, with what confidence and alternatives?

4 SLR methodology

The review follows the predefined Group B protocol, software-engineering review guidance and PRISMA reporting [4, 5, 6, 7]. Occurrence records retain their original database, run, file, position and hash. Deterministic duplicate links remain separate from possible versions. Preliminary title/abstract recommendations remain separate from final selection decisions. Relevant reports are assessed against all IC1-IC6 conditions. Study-level identity, report-level appraisal and individual outcome rows are distinct units.

The protocol plans independent screening and appraisal, with adjudication where needed. This checkpoint does not claim that those independent reviews have been completed. Existing human ACM work is preserved as existing work; it is not counted again as a new independent review. Only verified, record-keyed human votes can close the relevant validation fields.

5 Required review activities and current state

Activity	Measured state
Active imports	2,291 occurrences; 2,247 canonical records
First recommendations	2,247; 0 not yet recommended
Preliminary decisions	1,181 exclusion recommendations; 1,066 retain/uncertain
Version audit	41 pairs audited; proposals remain separate from applied duplicate links
Bounded source checks	5 priority reports; final eligibility pending
Quality assessment	50 item rows across 5 preliminary forms
Extraction	26 outcome rows; 5 study-level forms
Citation chasing	10 targeted events; not exhaustive
Final included studies	Pending; no final number established

6 Protocol and documented deviations

The submitted protocol dated 27 September 2026 is the governing baseline. Its 12 main-window database expressions, four older-MPS lanes, conjunctive eligibility criteria, Q1-Q10 rubric and extraction fields are preserved. Later execution differences are recorded separately.

Deviation	Handling and remaining consequence
Attributed shared searches	IEEE/Springer captures originated with Group A; they are reused by authorization, not claimed as newly executed Group B runs.
Fourth database deferred	ScienceDirect originals remain preserved outside active counts; four-database completion is not claimed.
ACM provenance partly recovered	Teammate log supplies an exact query, field, date and claimed counts; current exports do not reconcile those totals or the 79 numbered batches.
Older-MPS lanes	Prespecified historical searches have not been demonstrated complete; older citation leads require individual exceptions.
Independent validation incomplete	Initial recommendations and source checks do not establish independent consensus or final eligibility.

7 Academic database selection and provenance

Database selection follows the NED library resources specified in the protocol [8, 9].

Source	Occurrences	Source and coverage
ACM	1582	79 numbered supplied BibTeX batches; original run-to-file mapping unresolved
Springer	587	Attributed Group A full-text-indexed search captures
IEEE	102	Attributed Group A executed metadata captures
Supplementary	20	20 shared selected leads, separately identified

The active database-source total is 2,271 and the separate supplementary total is 20. The original raw manifest verifies 81 payload files. File hashes establish unchanged bytes; they do not establish that every search page was exported. Group A acceptance and its different QA scale are prior assessments rather than Group B final decisions.

8 Search strategy and executed queries

The exact Group B planned strings remain in the submitted protocol and query register. Imported executed queries are printed in Appendix A. Keyword overlap is not treated as proof of eligibility; the main search is intended to be sensitive, followed by criterion-based screening.

8.1 Recovered ACM search log

The teammate log reports: ACM Full-Text Collection; All Fields; 2020-2026; 14 October 2026 at 22:30 Pakistan time (UTC+05:00); 1,600 displayed hits and 1,603 claimed exported records. These are source-reported historical details, not a search re-executed in this cycle. The full-year filter can include publications after the actual search date, so publication date still needs verification.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
```

Recovered file	Parsed entries	SHA-256
ACM_MASTER_01.bib	829	a879137e9541e94e4e83ba6b90b330a7 / 05e902f7a7d6d2de783f62f23b5128a9
ACM_MASTER_02.bib	1000	54bfb9f1e1ac2b60e7fa2469135b0810 / 6593457f59cf7711cc5924fb125c137f

The recovered masters contain 1,829 overlapping occurrences. They yield 68 unmatched DOI/title candidate identifiers relative to the active payload. They are staged separately and not added to active identification counts. Existing numbered files are not automatically assigned to the recovered run. The claimed 1,603 total is not reproduced by these files. Full hashes and exact reconciliation are retained in the evidence registers.

9 Literature period and cutoff

The protocol main window is 1 January 2020 through the actual final run date in 2026, with individually justified older seminal exceptions. Shared IEEE/Springer captures use their documented 11 October 2026 search cutoff. The recovered ACM log reports 14 October 2026. These different dates are disclosed. Upload dates are not substituted for search dates, and an unverified publication year does not establish eligibility.

An older primary study requires a specific mechanism/baseline need, the same IC1-IC5 and QA checks, and a recorded exception decision. Older context-only papers remain separate. No studies outside a verified cutoff are silently treated as included.

10 Inclusion, exclusion and screening

Code	Rule
IC1	Final English peer-reviewed full conference/journal primary report.
IC2	Evaluated classifiable local sharing mechanism among concurrent tenants on one physical GPU.
IC3	Inference/model-serving workload; mixed training/inference results must be separable.
IC4	Inspectable relevant outcome with sufficient method/configuration for interpretation.
IC5	Extractable within-device result and sufficient available full text for appraisal/extraction.
IC6	Main window or documented seminal exception.
EC1	Secondary, non-primary/non-peer-reviewed, poster/abstract-only, preprint-only or container.
EC2	Training-only or no evaluable concurrent inference-sharing outcome.
EC3	Hardware/microarchitecture only without evaluated systems sharing mechanism.
EC4	Only aggregate cluster/remote outcomes, with no extractable within-GPU co-tenant result.
EC5	Non-GPU platform without relevant evaluated GPU mechanism.
EC6	Insufficient assessed full-text methods/configuration/outcomes.
EC7	Outside window without accepted exception, or non-English.

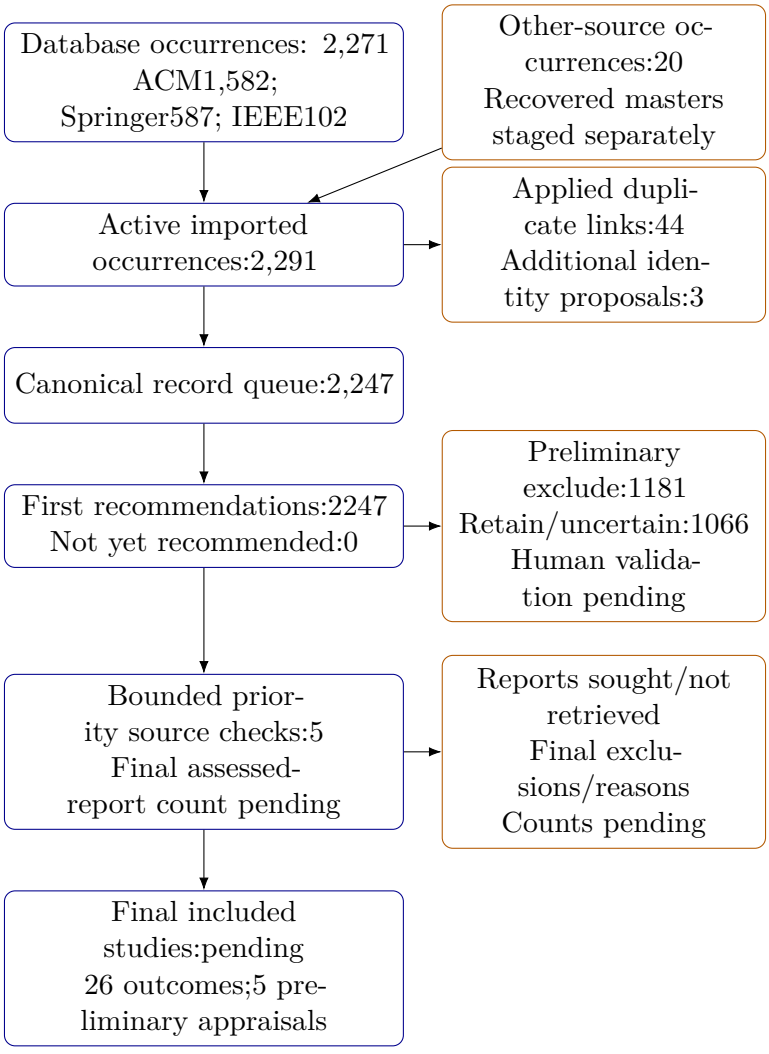
Title/abstract recommendations are preliminary. Ambiguity is retained; missing abstracts are recorded as abstract unavailable. Search-result snippets are not full abstracts. Recovered abstracts are connected through a DOI or a separately flagged title match. Missing full text is a retrieval status rather than EC6. Final exclusions require the applicable completed decision workflow.

Screening measure	Count
Existing preliminary recommendations preserved	800
New Cycle 1 recommendations	1447
Total first recommendations	2247
Preliminary exclusions	1181

Screening measure	Count
Retain/uncertain	1066
Not yet recommended	0
Final independent exclusions	Pending

11 PRISMA 2020 record accounting

The flow distinguishes identified occurrences, deterministic duplicate links, the current canonical queue and incomplete final selection stages. Preliminary recommendations do not populate final records-excluded or final included-study boxes. Proposed version/identity links have not changed active counts.



Possible-pair audit disposition	Pairs
companion_report_relationship_proposed	3
container_pair_scope_classified	30
identity_verified_linkage_proposed	3
unresolved_paper_identity	5

Three primary-metadata identity proposals and three companion-report proposals await reconciliation in the linkage register. Five paper-pair identities remain unresolved. Thirty whole-container pairs are scope-classified; they are not eligible study reports. No additional duplicate removal is applied in this cycle.

12 Search and screening logs

Each original occurrence retains an immutable record identifier and source membership. Recommendation events preserve stage, criterion, short reason, reviewer label, actual date/time when recorded and pending final state. Original historical timestamps are not reconstructed. Full-text access, quality and extraction checks retain separate source anchors.

Register	Traceability
records.csv / import_manifest.csv	Original occurrence metadata and original raw hashes.
screening.csv / title_votes.csv	Current preliminary queue and preserved recommendation events.
metadata_enrichment.csv	Recovered abstract metadata and match basis; raw records unchanged.
duplicates.csv / version_review_audit.csv	Applied duplicate links versus proposed identities and unresolved cases.
fulltext_checks.csv / cycle1_source_receipts.csv	Bounded eligibility checks and source PDF identities.
human_validation_queue.csv	All active retained/uncertain records, assigned for validation.
exclusion_validation_sample.csv	127 exclusions selected by source/criterion strata for checking.
snowballing.csv	10 citation events; no automatic new database count.

The exclusion sample uses deterministic record-ID ordering within source/criterion strata, taking 10% rounded up and at least two where available. It has not been marked checked. A false exclusion found during validation must trigger review of the relevant recommendation stratum.

12.1 Targeted citation events

Seed / direction	Candidate / evidence anchor
GB-P003 gpulets / backward	Prophet: Precise QoS Prediction on Non-Preemptive Accelerators; pp210-211 References [4]
GB-P003 gpulets / backward	Baymax: QoS Awareness and Increased Utilization for Non-Preemptive Accelerators; pp210-211 References [5]
GB-P003 gpulets / backward	Clipper: A Low-Latency Online Prediction Serving System; pp210-211 References [15]
GB-P003 gpulets / backward	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform; pp210-211 References [17]

Seed / direction	Candidate / evidence anchor
GB-P003 gpulets / backward	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up; p211 references[21]
GB-P003 gpulets / backward	Cocktail: A Multidimensional Optimization for Model Serving in Cloud; p211 references[22]
GB-P003 gpulets / backward	INFaaS: Automated Model-less Inference Serving; p212 references[36]
GB-P003 gpulets / backward	Nexus: A GPU Cluster Engine for Accelerating DNN-Based Video Analysis; p212 references[38]
GB-P005 REEF / backward	PipeSwitch: Fast Pipelined Context Switching for Deep Learning Applications; p555 references[6]
GB-P004 PipeSwitch / forward	Microsecond-scale Preemption for Concurrent GPU-accelerated DNN Inferences; REEF p555 references[6] verifies later paper cites PipeSwitch; bounded forward edge, not index census

A bounded forward edge from PipeSwitch to REEF is verified by REEF citing PipeSwitch in its bibliography. This proves that citation edge, not an exhaustive forward-citation search. Candidate occurrences already present are linked rather than counted again.

13 Quality assessment

The submitted Group B Q1-Q10 rubric scores 0 for absent/unclear, 1 for partial and 2 for sufficiently supported reporting. A justified N/A removes that item from the applicable maximum. Scores characterize reporting/internal validity, not venue prestige or an automatic inclusion gate. The below-60% sensitivity rule is retained for later synthesis; it is not applied to create a final study set.

Item	Meaning
Q1	Clarity of problem/objective
Q2	Mechanism and technical description
Q3	Experimental transparency and result provenance
Q4	Hardware/software/workload detail
Q5	Baseline fairness and comparability
Q6	Defined utilization/throughput/density metrics
Q7	Latency statistic, interval, load and method
Q8	Direct multi-tenant interference conditions
Q9	Defined and appropriately tested isolation boundary
Q10	Replication detail and reported limitations

Report	Raw/applicable	Normalized	Status
GB-P001	13 / 18	72.2%	Independent check pending
GB-P002	17 / 20	85.0%	Independent check pending
GB-P003	18 / 20	90.0%	Independent check pending
GB-P004	17 / 20	85.0%	Independent check pending

Report	Raw/applicable	Normalized	Status
GB-P005	17 / 20	85.0%	Independent check pending

Five preliminary forms are available. The older VastPipe and iGniter calibration forms are preserved; gpulets, PipeSwitch and REEF conference appraisals are fresh source-anchored additions. The 0-1 Group A scores are not copied or doubled. No independent checker or reconciled score has been invented.

13.1 GB-P001 appraisal detail

Item/score	Rationale	Source anchor
Q1 / 2	Objective and local GPU portability question explicit	sections1,6.7
Q2 / 2	Tuned MPS quotas and CU masking described	sections6.7-6.8
Q3 / 1	GPU-specific repeats and variance unresolved in conversion	sections6.7-6.9
Q4 / 1	GPU/backend named; precise GPU loads and versions incomplete	sections6.7-6.8
Q5 / 1	Same hardware and no/default/tuned MPS; quotas incomplete	Fig14
Q6 / 2	Throughput ratios and baseline named in prose	Fig14 and prose
Q7 / NA	GPU-portability extraction makes no latency claim; do not score NPU latency as GPU evidence	sections6.7-6.8
Q8 / 2	Multiple named co-located inference pairs evaluated	Figs14-16
Q9 / 1	Performance interference considered; no separate security/fault validation extracted	sections6.7-6.8
Q10 / 1	Stable-mix Q-table constraint; GPU replication settings incomplete	section6.9

13.2 GB-P002 appraisal detail

Item/score	Rationale	Source anchor
Q1 / 2	Interference-aware inference provisioning objective explicit	abstract, section1
Q2 / 2	MPS allocation and shadow implementation described	section4.2
Q3 / 2	Model checks repeated three times with SD; metric windows specified	sections5.1-5.3
Q4 / 1	GPU/workloads/backend detailed; OS/driver unresolved	sections4.2,5.1

Item/score	Rationale	Source anchor
Q5 / 1	Adapted baselines and allocation/batch differences disclosed	section5.1
Q6 / 2	Cost, throughput, resource-allocation metrics distinguished	sections5.1,5.3
Q7 / 2	P99 windows, SLO and recovery timing explicit	section5.3 Fig17
Q8 / 2	Four co-located models and quota configuration specified	section5.2 Fig13
Q9 / 1	Performance interference measured; security/ fault boundaries not tested in checked portion	sections5.2-5.3
Q10 / 2	Code, prediction limitations and PCIe caveat explicit	sections4.2,5.2

13.3 GB-P003 appraisal detail

Item/score	Rationale	Source anchor
Q1 / 2	Clear SLO-aware co-location objective	p199 abstract; section1
Q2 / 2	MPS partitions and temporal batching described	pp201-207 sections2-4
Q3 / 2	Poisson requests; median of3 repeats; throughput/SLO procedure	pp207-208 section5.1-5.2
Q4 / 2	Full GPU/CPU/OS/driver/runtime and model tables	p207 Tables2-3
Q5 / 2	Ported SBP comparator and Nexus incompatibility justified	p207 section5.1
Q6 / 2	Maximum throughput with99%SLO definition; system scale explicit	p208 section5.2
Q7 / 2	SLO99% procedure; local overhead CDF population explicit	p203 Fig7; p208 section5.2
Q8 / 2	Paired model interference and scheduler interference comparison	p203 Fig7; p208 Fig14
Q9 / 1	Performance interference checked; security/ fault boundary not extracted	p203 Fig7; section5.2
Q10 / 1	Prototype and appendix setup available; replication not attempted	p207 Table2; appendixA

13.4 GB-P004 appraisal detail

Item/score	Rationale	Source anchor
Q1 / 2	Fine-grained task-switching problem and design explicit	p499 abstract; section1
Q2 / 2	Controller, memory daemon, workers and pipeline described	pp504-508 sections4-5

Item/score	Rationale	Source anchor
Q3 / 2	100-run mean; min/max forFig6; setup described	p508 section6
Q4 / 1	Hardware/software specified; OS/kernel/ driver not extracted	p508 section6
Q5 / 2	Same modified PyTorch; ready-model and swapping conditions explicit	p508 section6.1
Q6 / 2	Inference throughput and utilization denominator defined	p509 section6.1 Fig6
Q7 / 2	End-to-end overhead defined; mean and extrema identified	pp508-509 Table3
Q8 / 2	Training/inference co-location; inference metric separated	pp508-509 section6.1
Q9 / 1	Process/memory control described; security/ fault testing not extracted	pp507-508 implementation
Q10 / 1	Versioned stack described; limitations documented; replication not attempted	p508 section6; section7

13.5 GB-P005 appraisal detail

Item/score	Rationale	Source anchor
Q1 / 2	Concurrent real-time/best-effort inference objective explicit	p539 abstract; section1
Q2 / 2	Reset preemption, queues and padding implementation described	sections3-6; pp547-548
Q3 / 1	Benchmark arrivals and baseline methods detailed; repeat/variance protocol not extracted	p548 section7.1
Q4 / 2	CPU, GPU, host RAM, OS, runtime, models andclients specified	p548 section7.1 Table2
Q5 / 2	RT-only and GPUStreams/SEQ conditions; extensions disclosed	p548 section7.1; p549 section7.3
Q6 / 2	Total RT+BE throughput and denominator explicit	p549 section7.2 Fig11
Q7 / 1	End-to-end latency and load specified; request percentile not extracted	pp548-549 Table2/Fig11
Q8 / 2	MultipleRT/BE clients and multiple mixes directly tested	p548 Table2; p549 section7.2
Q9 / 1	Performance isolation tested; no extracted memory/security/fault boundary test	pp548-549 section7.2
Q10 / 2	Artifact and idempotence/platform limitations explicit	p540 repository; section8; appendixA

14 Data extraction

The extraction register keeps each outcome attached to a report, mechanism, hardware, workload, baseline, metric, value/unit, source anchor and caveat. Study forms preserve author/year/venue, CPU/memory/software, OS/driver/runtime, model/tenant mix, load/SLO, resource use, scalability, energy, isolation, overhead, advantages, limitations and trade-offs. Rows are observations, not independent studies.

Use not extracted for an unchecked field. Use not reported only after checking the source sufficiently to establish absence. No zero is substituted for missing data. Exact author-reported values are transcribed; no graph digitization or numerical interpolation was performed. All headline values remain assigned for independent verification.

Metric safeguards: additional latency is not total latency; an across-model-pair CDF is not a request-latency percentile; prediction error is not speedup; total RT-plus-BE throughput is not real-time-only throughput; and performance interference does not establish security or fault containment. Multi-GPU scheduling benefits are separated from local GPU outcomes.

Primary source identities for the checked reports: [10, 11, 12, 13, 14].

14.1 GB-P001 VastPipe

Field	Source-checked entry / remaining field
Bibliographic identity	Li-Xian Ma and Le-Ping Wang and En Shao and Rong-Yu Cao and Guang-Ming Tan; 2025; Journal of Computer Science and Technology; 10.1007/s11390-024-3773-5
Mechanism	tuned/default MPS comparison; prior source check only
GPU	NVIDIA A100 (existing source-check register)
CPU and memory	not extracted
OS / kernel	not extracted
Driver / runtime	not extracted
Models and workload	VGG16+GoogLeNet (existing extraction rows)
Tenants	not extracted
Batch / load / SLO	not extracted
Baseline	No MPS / default MPS; exact quotas not extracted
Throughput	not extracted
Latency	not extracted
Resource usage	not extracted
Scalability	not extracted
Energy	not extracted
Reliability	not extracted
Security / fault isolation	not extracted
Overhead	not extracted
Advantages	not extracted
Limitations	not extracted
Trade-offs	not extracted
Source anchor	pp457-459, sections6.7-6.9, Figs14-16

Outcome observations

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
VGG16+GoogLeNet; MPS tuned quotas	Throughput ratio: 2.7 x	No MPS; NVIDIA A100	pp457-458 Fig14 and accompanying prose; Batch/rate/ quotas unresolved; not tail-latency evidence
VGG16+GoogLeNet; MPS tuned quotas	Throughput ratio: 1.5 x	Default MPS; NVIDIA A100	pp457-458 Fig14 and accompanying prose; Same comparison family; not a universal average

14.2 GB-P002 iGniter

Field	Source-checked entry / remaining field
Bibliographic identity	Fei Xu and Jianian Xu and Jiabin Chen and Li Chen and Ruitao Shang and Zhi Zhou and Fangming Liu; 2023; IEEE Transactions on Parallel and Distributed Systems; 10.1109/ TPDS.2022.3232715
Mechanism	MPS quota allocation and shadow process recovery; prior source check
GPU	NVIDIA V100 (existing source-check register)
CPU and memory	not extracted
OS / kernel	not extracted
Driver / runtime	not extracted
Models and workload	AlexNet; ResNet50; VGG19; SSD
Tenants	not extracted
Batch / load / SLO	batch3 for prediction check; W1 SLO10ms for recovery example
Baseline	Measured latency for prediction error;10ms SLO for recovery
Throughput	not extracted
Latency	not extracted
Resource usage	not extracted
Scalability	not extracted
Energy	not extracted
Reliability	not extracted
Security / fault isolation	not extracted
Overhead	not extracted
Advantages	not extracted
Limitations	not extracted
Trade-offs	not extracted
Source anchor	pp820-824, sections4.2,5.1-5.4, Figs13-17

Outcome observations

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
AlexNet/ResNet50/ VGG19/SSD; batch3; MPS 25% per model	Prediction error range: 1.53-5.02 percent	Measured local latency; NVIDIA V100	p822 section5.2 Fig13; Prediction accuracy, not utilization or latency speedup
W1 with W5 and W11; MPS shadow-process recovery	Initial p99 latency: 15.6 ms	W1 latency SLO10ms; NVIDIA V100	p823 section5.3 Fig17 prose; Transient violation precedes recovery; no unconditional guarantee

14.3 GB-P003 gpulets

Field	Source-checked entry / remaining field
Bibliographic identity	Seungbeom Choi et al.; 2022; USENIX ATC22; not resolved
Mechanism	MPS spatial partitions plus temporal batching; interference-aware gpulet scheduler
GPU	RTX2080Ti11GB; two GPUs/server, two servers; local pair measurements separated
CPU and memory	20-core Xeon E5-2630v4;192GB DRAM (Table2)
OS / kernel	Ubuntu18.04; kernel not extracted
Driver / runtime	NVIDIA440.64; CUDA10.2; PyTorch1.10 (p207 Table2)
Models and workload	GoogLeNet, LeNet, ResNet50, SSD-MobileNet, VGG16, MnasNet, MobileNetv2, DenseNet, BaseBERT (Table3)
Tenants	paired local inferences; system scenarios contain several models
Batch / load / SLO	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
Baseline	Solo latency for local Fig7; SBP and greedy best-fit for system scheduling
Throughput	not extracted
Latency	Fig7 across-pair CDF threshold17.66%; not request-p90
Resource usage	not extracted
Scalability	8 logical backend servers with additional GPUs; system-level result (pp208-209)
Energy	not extracted
Reliability	not extracted
Security / fault isolation	not extracted
Overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
Advantages	not extracted
Limitations	Original Nexus not directly comparable; SBP ported; cross-model interference remains; no extracted security/fault test

Field	Source-checked entry / remaining field
Trade-offs	Partition and batch jointly selected; throughput numbers at system scale cannot be assigned to one GPU
Source anchor	pp201-203 local mechanism/Fig7; p207 Tables2-3; pp208-209 system methodology

Outcome observations

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
Paired model inferences; MPS spatial colocation	Distribution threshold:90% of measured pairs: 17.66 percent overhead below threshold	Solo-running latency; RTX2080Ti11GB	p203 Fig7; Across-pair CDF, not request-p90/p99; no hostile-memory/fault test

14.4 GB-P004 PipeSwitch

Field	Source-checked entry / remaining field
Bibliographic identity	Zhihao Bai; Zhen Zhang; Yibo Zhu; Xin Jin; 2020; USENIX OSDI20; not resolved
Mechanism	Temporal task switching; pipelined model transfer; active/standby workers and shared memory daemon
GPU	V10016GB PCIe3x16; T416GB PCIe3x8
CPU and memory	V100 instance:8vCPU Xeon E5-2686v4/61GB; T4 instance:8vCPU Platinum8259CL/32GB (p508)
OS / kernel	not extracted
Driver / runtime	CUDA10.1; PyTorch1.3.0; torchvision0.4.2; scipy1.3.2; driver not extracted
Models and workload	ResNet152; Inception_v3; Bert_base
Tenants	one training task preempted for one inference task; separable inference latency
Batch / load / SLO	training32; inference8;100-run averages; Fig6 reports minimum/maximum (p508)
Baseline	Ready model; stop-and-start; MPS using unified memory
Throughput	not extracted
Latency	Additional total latency versus ready model; Table3; not p99
Resource usage	not extracted
Scalability	not extracted
Energy	not extracted
Reliability	not extracted
Security / fault isolation	not extracted

Field	Source-checked entry / remaining field
Overhead	Table3 total extra latency; Table4 first-layer startup overhead; kept separate
Advantages	Reduced switching overhead for checked workloads; see extraction.csv
Limitations	Mixed training/inference applicability; MPS benchmark swaps memory; modified PyTorch for all comparators; no extracted OS/driver/security/fault evidence
Trade-offs	Pipelined transfers and registered resident host models require runtime integration; inference-only multi-tenant relevance needs human eligibility check
Source anchor	p508 section6/6.1; p509 Tables3-4

Outcome observations

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
ResNet152 inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 6.01 ms	Ready-model latency subtracted; MPS overhead307.02ms; NVIDIA V10016GB	p509 Table3; setup p508 section6; Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
ResNet152 inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 307.02 ms	Ready-model latency subtracted; NVIDIA V10016GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Inception_v3 inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 5.40 ms	Ready-model latency subtracted; NVIDIA V10016GB	p509 Table3; setup p508 section6; Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Inception_v3 inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 232.25 ms	Ready-model latency subtracted; NVIDIA V10016GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
Bert_base inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 10.27 ms	Ready-model latency subtracted; NVIDIA V10016GB	p509 Table3; setup p508 section6; Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Bert_base inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 204.52 ms	Ready-model latency subtracted; NVIDIA V10016GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
ResNet152 inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 5.57 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
ResNet152 inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 259.20 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Inception_v3 inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 7.66 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Inception_v3 inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 193.05 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Bert_base inference batch8 + training batch32; Temporal switching	Additional end-to-end latency: 34.56 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
Bert_base inference batch8 + training batch32; NVIDIA MPS with unified-memory swapping	Additional end-to-end latency: 338.25 ms	Ready-model latency subtracted; NVIDIA T416GB	p509 Table3; setup p508 section6; Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.

14.5 GB-P005 REEF conference

Field	Source-checked entry / remaining field
Bibliographic identity	Mingcong Han; Hanze Zhang; Rong Chen; Haibo Chen; 2022; USENIX OSDI22; not resolved
Mechanism	Reset-based preemption; idempotent inference kernels; dynamic kernel padding; runtime/driver changes
GPU	AMD Radeon Instinct MI50:60CUs/16GB; REEF-N extension NVIDIA V100
CPU and memory	Intel Corei7-10700:8cores;16GB host DRAM (p548)
OS / kernel	Ubuntu18.04; kernel not extracted
Driver / runtime	ROCm4.3.0; ApacheTVM0.8.0; NVIDIA extension CUDA10.2
Models and workload	ResNet152; DenseNet201; VGG19; Inceptionv3; DistilBERT
Tenants	DISB A/B:1RT+1BE; C:1RT+5BE; D/E:5RT+5BE
Batch / load / SLO	Table2 RT100/220/100/20/20req/s; D uniform, E Poisson; BE closed-loop; batch not extracted
Baseline	RT-Only; SEQ; GPUStreams; separate wait-based preemption microbenchmark
Throughput	RT+BE total normalized to RT-Only; not RT-only throughput
Latency	RT end-to-end overhead below2% in REAL; no request percentile extracted
Resource usage	not extracted
Scalability	not extracted
Energy	not extracted
Reliability	not extracted
Security / fault isolation	not extracted
Overhead	not extracted
Advantages	Checked workloads retain near RT-only latency while using idle capacity for BE inference
Limitations	Reset requires idempotent kernels; model-pair padding incompatibilities; conference/journal report linkage open; not security/fault isolation

Field	Source-checked entry / remaining field
Trade-offs	DISB D/E throughput about25% below GPUStreams while limiting RT latency; implementation differs on closed-source GPUs
Source anchor	p548 section7.1 Table2; p549 section7.2 Fig11; section8 idempotence limitation

Outcome observations

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE; Reset-based kernel preemption + dynamic kernel padding	Overall throughput ratio: 1.60 x	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE; Reset-based kernel preemption + dynamic kernel padding	RT end-to-end latency overhead: 0.5 percent	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; Described as negligible; not a p99 value or worst-case guarantee.
DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE; Reset-based kernel preemption + dynamic kernel padding	Overall throughput ratio: 1.14 x	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; High RT load; differs from low-load DISB A.
DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE; Reset-based kernel preemption + dynamic kernel padding	RT end-to-end latency overhead: 1 percent	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; About60 microseconds in prose; not a tail percentile.
DISB C-E:multiple BE clients; Reset-based kernel preemption + dynamic kernel padding	RT end-to-end latency overhead: <1.5 percent	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; Authors also state0.1ms; keep inequality and scenario range.
DISB D:5 RT model clients20req/s uniform +5 BE model clients; Reset-based kernel preemption + dynamic kernel padding	Overall throughput ratio: 3.00 x	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; About25% lower throughput than GPUStreams; latency trade-off preserved.

Workload / mechanism	Metric / value	Baseline / hardware	Anchor / caveat
DISB E:5 RT model clients20req/s Poisson +5 BE model clients; Reset-based kernel preemption + dynamic kernel padding	Overall throughput ratio: 2.96 x	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; About25% lower throughput than GPUStreams; latency trade-off preserved.
Apollo REAL trace about43 RTreq/s +5 BE model clients; Reset-based kernel preemption + dynamic kernel padding	Overall throughput ratio: 7.7 x	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; Real-world trace mapped to representative DNN models; not production deployment measurement.
Apollo REAL trace about43 RTreq/s +5 BE model clients; Reset-based kernel preemption + dynamic kernel padding	RT end-to-end latency overhead: <2 percent	RT-Only dedicated GPU; AMD MI5016GB	p549 section7.2 Fig11 prose; setup p548 Table2; Not a p99 latency claim or security/fault isolation test.

A Exact attributed shared executed searches

A.1 IEEE-MPS

Database: IEEE Xplore. Date: 2026-10-11; recorded UTC time: not_recorded. Originating operator: AI-CODEX. Collection: Xplore all metadata.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MPS OR "multi-process service")
```

Field: All metadata (Search by All); not field-qualified Command Search. Filters: Publication years2020-2026; date/language/type checked during screening. Displayed hits: 25. Captured occurrences: 25. Method: Public rendered-result metadata; all pages captured; publisher CSV requires personal sign-in; BibTeX export timed out.

A.2 IEEE-MIG

Database: IEEE Xplore. Date: 2026-10-11; recorded UTC time: not_recorded. Originating operator: AI-CODEX. Collection: Xplore all metadata.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
```

Field: All metadata (Search by All); not field-qualified Command Search. Filters: Publication years2020-2026; date/language/type checked during screening. Displayed hits: 23. Captured occurrences: 23. Method: Public rendered-result metadata; all pages captured; publisher CSV requires personal sign-in; BibTeX export timed out.

A.3 IEEE-TEMPORAL

Database: IEEE Xplore. Date: 2026-10-11; recorded UTC time: not_recorded. Originating operator: AI-CODEX. Collection: Xplore all metadata.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching")
```

Field: All metadata (Search by All); not field-qualified Command Search. Filters: Publication years2020-2026; date/language/type checked during screening. Displayed hits: 45. Captured occurrences: 45. Method: Public rendered-result metadata; all pages captured; publisher CSV requires personal sign-in; BibTeX export timed out.

A.4 IEEE-API

Database: IEEE Xplore. Date: 2026-10-11; recorded UTC time: not_recorded. Originating operator: AI-CODEX. Collection: Xplore all metadata.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
```

Field: All metadata (Search by All); not field-qualified Command Search. Filters: Publication years2020-2026; date/language/type checked during screening. Displayed hits: 9. Captured occurrences: 9. Method: Public rendered-result metadata; all pages captured; publisher CSV requires personal sign-in; BibTeX export timed out.

A.5 SPR-MASTER

Database: Springer Nature Link. Date: 2026-10-11; recorded UTC time: not_recorded. Originating operator: AI-CODEX. Collection: Springer Nature Link.

```
(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
```

Field: General search/full-text indexed collection. Filters: English; Research article OR Conference paper;2020-2026; manual cutoff. Displayed hits: 587. Captured occurrences: 587. Method: Rendered metadata from all30 pages; publisher CSV routes to sign-in.

B Validation and continuation

Register snapshot generated 2026-10-15T22:22:20+00:00. Original 81-file import hashes and submitted protocol bytes pass consistency checks. Final included-study count remains pending. First-recommendation coverage is 100.0%; this is not final screening completion.

Assigned person	Exact next action
Eshaal	Reuse existing ACM human work; supply record-keyed decisions. Reconcile recovered query log to79 numbered exports and explain1600/1603 claims against829+1000 recovered entries. Check assigned uncertain records and the exclusion sample.
Rameen	Independently verify all5 priority reports,26 outcome rows and50 Q1-Q10 items using exact anchors. Preserve own scores/votes before comparing. Resolve mixed-workload eligibility and preferred REEF report.
Raazia	Import verified human decisions, reconcile3 identity proposals and3 companion relationships, resolve5 remaining paper pairs, then expand full-text eligibility/QA/extraction beyond the5 bounded reports.

Planning readiness is approximately 50-55%, with substantial uncertainty. Collection, a complete preliminary pass and traceable first-review evidence are established; independent validation, broader full-text extraction, search reconciliation and final PRISMA remain open. This is not a verified completion percentage or marks prediction. Later CEP taxonomy, synthesis, comparisons and engineering recommendation are reserved for Cycle 2.

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